

Estimation of Software

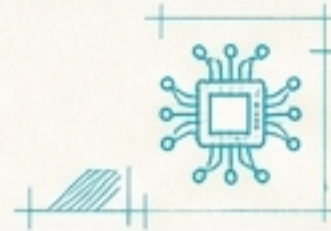
A framework for calculating project effort, time, and cost.

Software Project Management | Project Estimation

THE ANATOMY OF SOFTWARE PROJECT COSTS

DIRECT PROJECT COSTS

HARDWARE AND SOFTWARE COSTS



TRAVEL AND TRAINING COSTS



EFFORT COSTS (THE SALARIES OF ENGINEERS INVOLVED IN THE PROJECT)



INDIRECT / OVERHEAD COSTS

SOCIAL AND INSURANCE COSTS



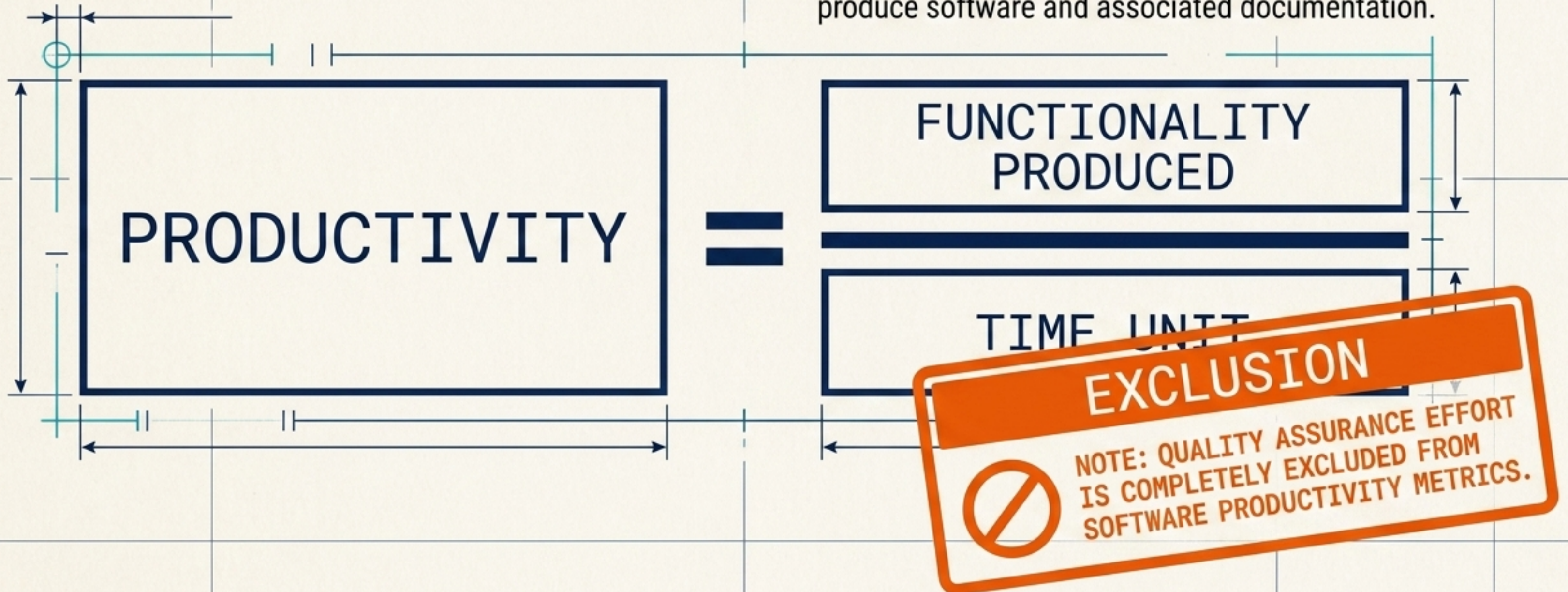
OVERHEADS (COSTS OF NETWORKING AND COMMUNICATIONS)



ESTIMATES DISCOVER THE COST OF PRODUCTION. THERE IS NOT A SIMPLE RELATIONSHIP BETWEEN THE DEVELOPMENT COST AND THE FINAL PRICE CHARGED TO THE CUSTOMER.

DEFINING SOFTWARE PRODUCTIVITY

Measures the rate at which developers produce software and associated documentation.

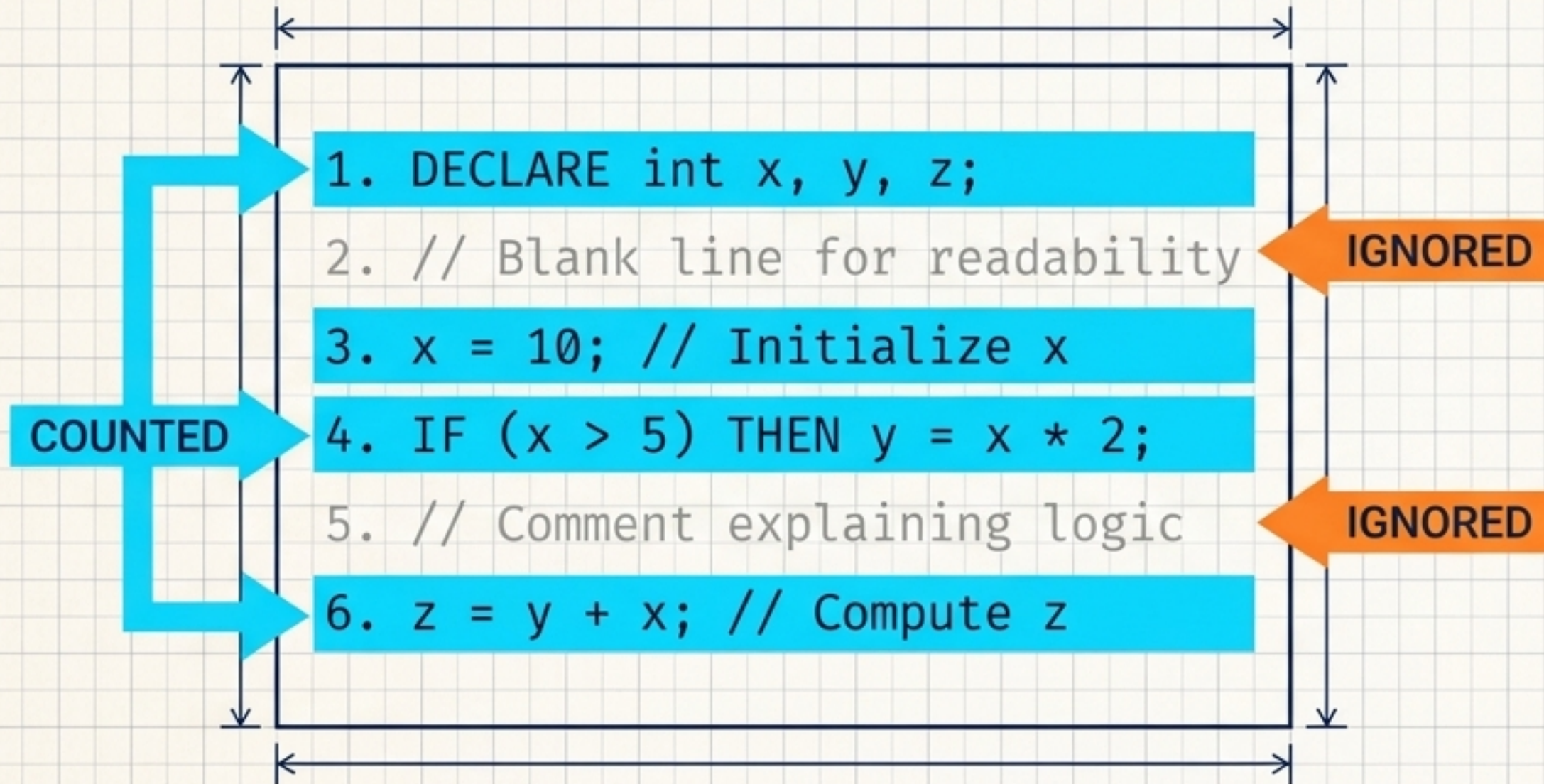


Two Paths to Estimation

Approach 1: Lines of Code (LOC)	Approach 2: Function Points (FP)
Core Philosophy: Size-related. Assesses based on project lines of source code.	Core Philosophy: Functionality-related. Assesses based on functionalities delivered.
Perspective: Developer-centric.	Perspective: User-centric (what they receive or enter).
Language Dependence: Highly dependent on the programming language used.	Language Dependence: Completely language-independent.
Metric: Measured in LOC / pm (person-months) or LOC / pm^6 .	

Approach 1: Lines of Code (LOC) Estimation

The Anatomy of a Count



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Advantage:

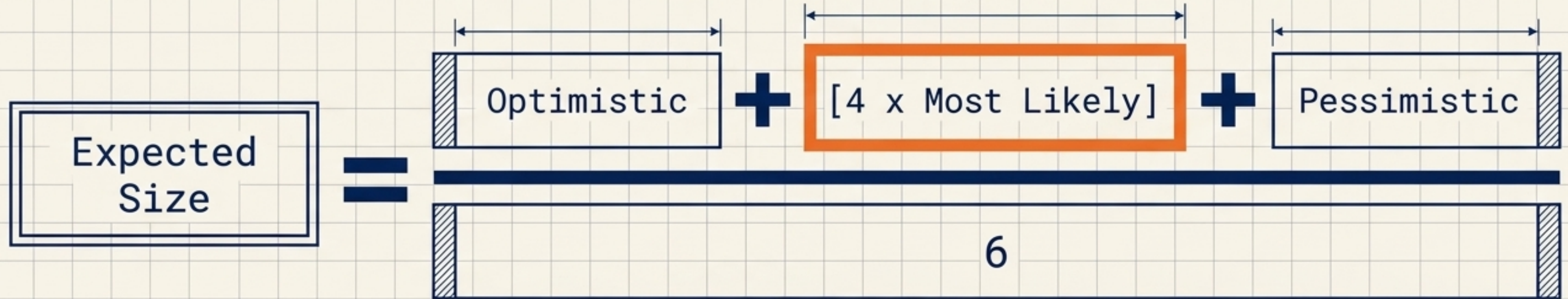
Extremely easy to count and calculate.

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Disadvantage:

Flawed metric due to its high dependency on the specific programming language chosen.

The 3-Point LOC Estimation Formula



Data Processing

Inputs

Optimistic = 4600 | Most Likely = 6900 | Pessimistic = 8600



Calculation flow

$(4600 + [4 * 6900] + 8600) / 6$



Output

6800 LOC

LOC Synthesis: From Code to Cost

Step 1: Size (Input)

Login:	1000
Search:	3000
Payment:	2000
Profile:	1500
Feedback:	500
Total =	8000 LOC

Step 2: Effort (Process)

$$\text{Effort} = \frac{\text{Total LOC}}{\text{LOC per person-month}}$$

Org. Avg: 600 LOC/pm

$$\frac{8000}{600} = 13.33 \text{ Person-Months}$$

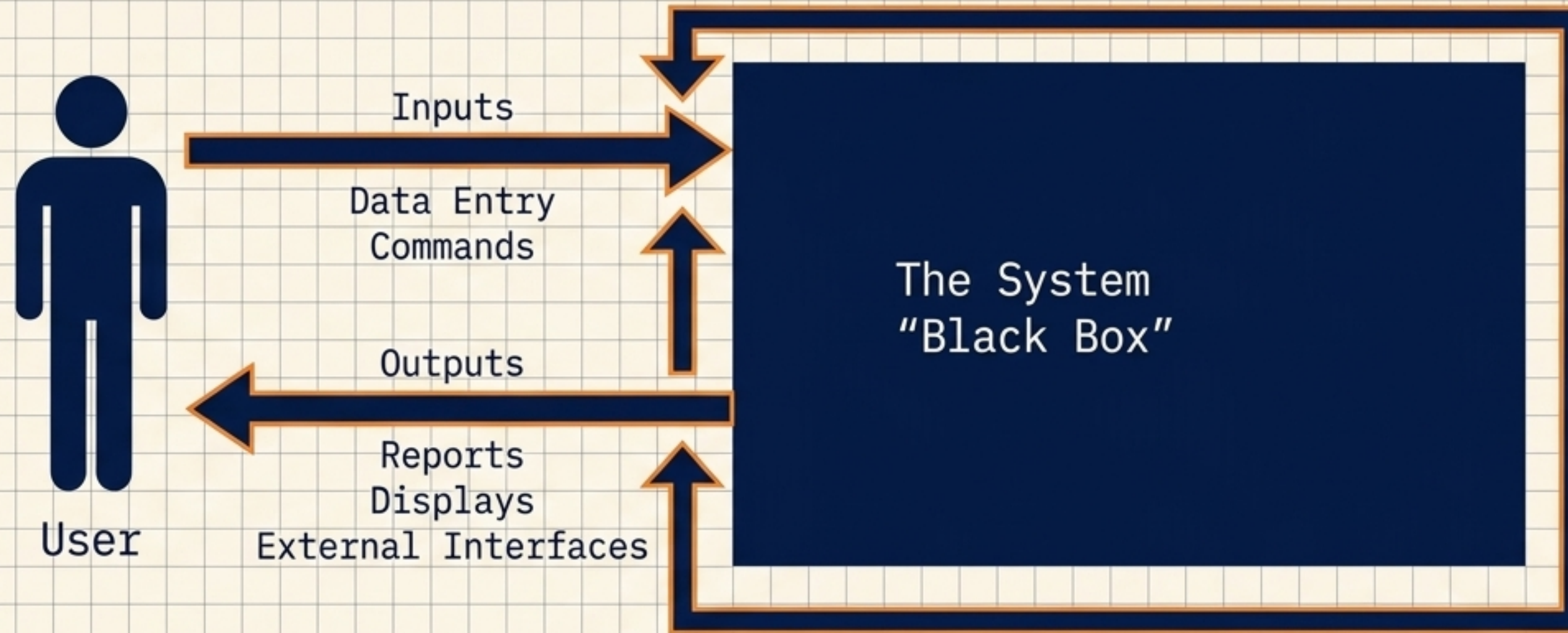
Step 3: Cost (Output)

$$\text{Cost} = \text{Total LOC} * \text{Cost per LOC}$$

Avg. cost: \$8000/month -> \$13.3/LOC

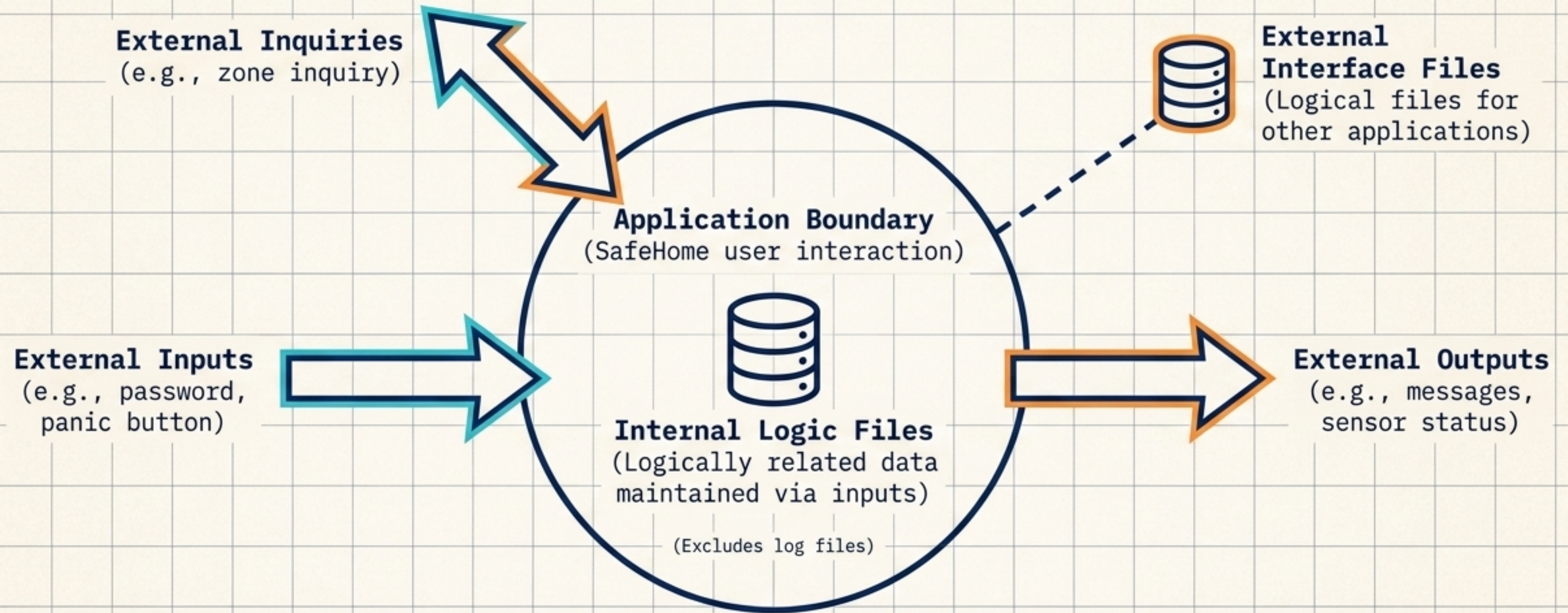
$$\text{\$13.3} * 8000 = \text{\$106,640}$$

Approach 2: Function Point (FP) Estimation



Function Point Estimation measures functionality strictly from a user's point of view——what is received from or entered into the system. It estimates inputs, outputs, inquiries, files, and external interfaces, applying weights based on the importance and complexity of the operation.

The 5 Pillars of Function Points



The FP Master Equation

$$FP = \boxed{UFP} * \boxed{CAF}$$

Unadjusted Functional Points

Defined as the baseline count of the 5 parameters, ranked and weighted by predefined complexity (Low, Average, High).

Complexity Adjustment Factor

Defined as an environmental multiplier based on the answers to 14 specific system design questions.

Calculating Unadjusted Function Points (UFP)

Parameter	Counts	Low	Average	High	Total
External Inputs	3	3	4	6	9
External Outputs	2	4	5	7	8
External Inquiries	2	3	4	6	6
Internal Logic Files	1	7	10	15	7
External Interfaces	4	5	7	10	20

Count Total (UFP) = 50

Calculating the Complexity Adjustment Factor (CAF)

Value Adjustment Questions (Fi)

[PERFORMANCE & RELIABILITY]

- ☐ Reliable backup/recovery required
- ☐ Specialized data communications
- ☐ Distributed processing functions
- ☐ Performance critical

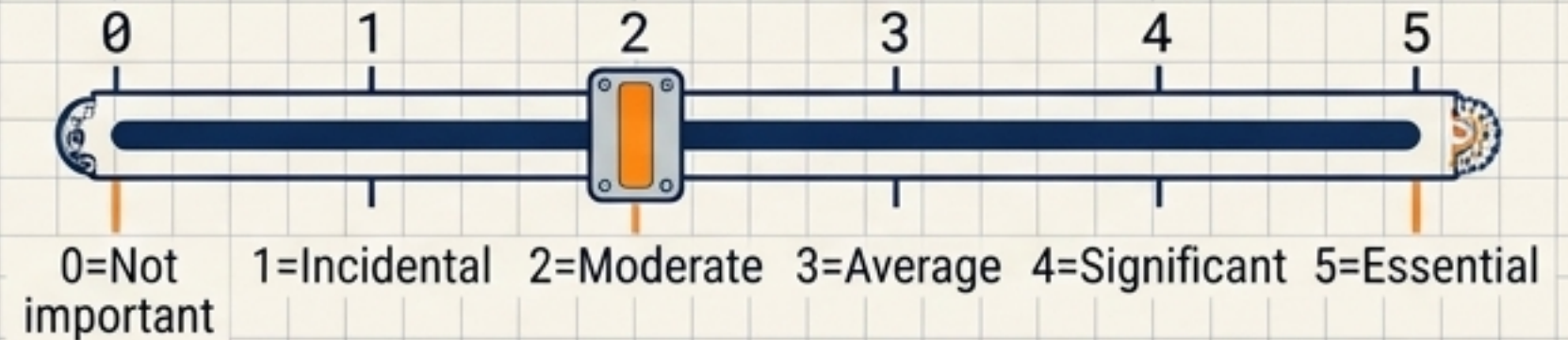
[OPERATIONS & INTERFACE]

- ☐ Heavily utilized operational environment
- ☐ Online data entry required
- ☐ Complex online data entry (multi-screen)
- ☐ ILFs updated online
- ☐ Complex inputs/outputs/files/inquiries
- ☐ Complex internal processing

[PROCESSING & REUSABILITY]

- ☐ Reusable code design
- ☐ Conversion/installation included
- ☐ Multiple installation design
- ☐ Change/ease of use facilitation

Complexity Scale (0-5)



$$CAF = 0.65 + (0.01 * \sum Fi)$$

[EXAMPLE CALCULATION]

If a product is moderately complex, all 14 questions are answered as 'Moderate' (2).

Total $\sum Fi = 14 * 2 = 28$.

Therefore, $CAF = 0.65 + (0.01 * 28) = 0.93$.

FP Synthesis: Establishing Project Duration

Total Function Points

$$\text{UFP (50)} * \text{CAF (0.93)} = 46.5 \text{ Total FP}$$

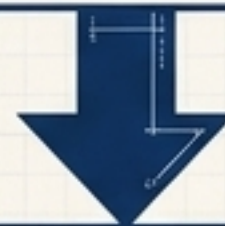


(Historical Productivity: 60 FP/person-week)

Total Effort

$$\text{Effort} = \text{FP} / \text{Productivity}$$

$$\text{Calculation: } 46.5 / 60 = 0.775 \text{ Person-Weeks}$$



(Team Size = 5)

Project Duration

$$\text{Duration} = \text{Effort} / \text{Team Size}$$

$$\text{Calculation: } 0.775 \text{ Person-Weeks} / 5 = \text{Final Project Duration}$$

Final Summary: Estimation Baseline Checklist

The Historical Data Imperative

- ✓ Organizational productivity rates (LOC/pm or FP/week)
- ✓ Personnel payment averages (\$/month)
- ✓ Complexity weighting profiles

Neither LOC nor FP methodologies can project real business costs without previously collected organizational baseline data.

The Core Takeaway

Software estimation is **never** just counting code or features. It is the process of **using baseline metrics collected from past projects** in conjunction with present **functionality** variables to **project accurate, real-world cost and effort.**